

A2 LDA Python - Answers

Python -Based LDA MCQs

1. Which Python library provides a built -in implementation of Linear Discriminant Analysis?

- ☐ A) NumPy
- ☐ B) Pandas
- ☐ C) Scikit -learn
- ☐ D) Matplotlib

Answer: C) Scikit-learn provides a built-in LinearDiscriminantAnalysis estimator.

2. Which class is commonly used to implement LDA in Scikit -learn?

- ☐ A) PCA
- ☐ B) LinearDiscriminantAnalysis
- ☐ C) LogisticRegression
- ☐ D) DecisionTreeClassifier

Answer: B) LinearDiscriminantAnalysis.

3. What is the purpose of the fit() method in Scikit -learn's LDA?

- ☐ A) Visualize data
- ☐ B) Learn discriminant directions from labeled data
- ☐ C) Normalize features
- ☐ D) Predict labels

Answer: B) fit() estimates the discriminant model from labeled training data.

4. Which method is used to reduce dimensionality after fitting an LDA model?

- ☐ A) transform()
- ☐ B) predict()
- ☐ C) score()
- ☐ D) classify()

Answer: A) transform() projects samples into the learned discriminant space.

5. In Python, which library is most suitable for computing eigenvalues and eigenvectors manually?

- ☐ A) OpenCV
- ☐ B) NumPy
- ☐ C) Flask
- ☐ D) Seaborn

Answer: B) NumPy provides matrix and eigenvalue operations.

6. What will happen if the target labels y are not provided while fitting an LDA model?

- ☐ A) LDA will automatically cluster data
- ☐ B) LDA cannot be trained properly

- o C) LDA converts to PCA
- o D) No effect

Answer: B) LDA is supervised; fitting requires target labels.

7. Which command is used to import LDA from Scikit -learn?

- o A) from sklearn.decomposition import LDA
- o B) from sklearn.lda import LDA
- o C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
- o D) from sklearn.linear_model import LDA

Answer: C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis.

8. What is the output shape after applying LDA with n_components=2 to a dataset containing 100 samples?

- o A) (2,100)
- o B) (100,2)
- o C) (100,100)
- o D) (2,2)

Answer: B) With 100 samples and two output components, the transformed shape is (100, 2), assuming two components are allowed.

9. Which NumPy function is commonly used to compute eigenvalues and eigenvectors?

- o A) np.mean()
- o B) np.linalg.eig()
- o C) np.dot()
- o D) np.var()

Answer: B) np.linalg.eig() (or eig() for a symmetric matrix).

10. In an LDA workflow, why is feature scaling often performed before applying LDA?

- o A) To reduce memory usage
- o B) To ensure features contribute comparably
- o C) To increase sample size
- o D) To remove labels

Answer: B) Scaling can help features on different numeric scales contribute comparably; it is not a universal requirement for every LDA solver.

11. Which parameter controls the number of discriminant components in Scikit -learn LDA?

- o A) n_clusters
- o B) n_neighbors
- o C) n_components
- o D) max_depth

Answer: C) n_components.

12. Which matrix operation is required when implementing LDA from scratch?

- o A) Matrix inversion

- o B) Image convolution
- o C) FFT
- o D) Hashing

Answer: A) Matrix inversion (or an equivalent linear solve) appears in the standard $S_W^{-1} S_B$ formulation.

13. While implementing LDA manually, which scatter matrix represents class compactness?

- o A) Between -Class Scatter Matrix
- o B) Within -Class Scatter Matrix
- o C) Covariance Matrix
- o D) Correlation Matrix

Answer: B) Within-class scatter measures compactness within each class.

14. Which step comes first when implementing LDA from scratch?

- o A) Compute eigenvectors
- o B) Compute class means
- o C) Train classifier
- o D) Plot decision boundary

Answer: B) Compute class means before constructing the scatter matrices.

15. After computing eigenvectors in manual LDA, what should be done next?

- o A) Randomly shuffle them
- o B) Sort them according to eigenvalues
- o C) Delete negative values
- o D) Normalize labels

Answer: B) Sort eigenvectors by descending eigenvalue to select the strongest discriminant directions.

16. Which NumPy function can be used to project data onto selected discriminant vectors?

- o A) `np.dot()`
- o B) `np.max()`
- o C) `np.sum()`
- o D) `np.argmax()`

Answer: A) `np.dot()` (or the `@` operator) projects samples onto selected vectors.

17. What is the primary purpose of `lda.predict(X_test)`?

- o A) Reduce dimensions
- o B) Classify unseen samples
- o C) Scale data
- o D) Compute covariance

Answer: B) `predict()` classifies unseen samples.

18. Which metric is commonly used to evaluate an LDA classifier?

- o A) Accuracy
- o B) RMSE

- o C) MAPE
- o D) MAE

Answer: A) Accuracy is a common classification metric, though other metrics may be preferable for imbalanced classes.

19. In Scikit -learn, LDA can be used for:

- o A) Classification only
- o B) Dimensionality reduction only
- o C) Both classification and dimensionality reduction
- o D) Clustering only

Answer: C) LDA supports both classification and dimensionality reduction.

21. You are asked to implement Linear Discriminant Analysis from scratch using only NumPy

(without using `sklearn.discriminant_analysis.LinearDiscriminantAnalysis`). Which sequence of steps is correct?

- A) Standardize Data → Compute Class Means → Compute Within -Class Scatter Matrix → Compute Between -Class Scatter Matrix → Solve Eigenvalue Problem of $S_W^{-1}S_B$ → Select Top Eigenvectors → Project Data
- B) Compute FFT → Compute Covariance Matrix → Apply K -Means → Project Data
- C) Compute PCA Components → Apply Logistic Regression → Predict Labels
- D) Normalize Labels → Compute Decision Tree → Extract Features → Project Data

The supplied PDF jumps from Question 19 to Question 21; no Question 20 appears in the source.

Answer: A) Standardize as appropriate, calculate class means and S_W/S_B , solve the generalized eigenproblem, sort/select vectors, then project.

22. What is the purpose of the following line in a manual LDA implementation?

`eigvals, eigvecs = np.linalg.eig(np.linalg.inv(Sw).dot(Sb))`

- A) Compute principal components
- B) Compute discriminant directions
- C) Compute cluster centroids
- D) Compute feature scaling factors

Answer: B) It computes eigenvalues and discriminant directions of $S_W^{-1} S_B$.

23. Consider:

`X_lda = X.dot(W)`

What does W represent?

- A) Weight initialization matrix
- B) Matrix of selected discriminant vectors
- C) Covariance matrix
- D) Distance matrix

Answer: B) W contains the selected discriminant vectors as columns.

24. In a scratch implementation, why are eigenvectors sorted in descending order of eigenvalues?

- A) To reduce execution time
- B) To select directions with maximum discriminative power
- C) To normalize data
- D) To balance classes

Answer: B) Descending eigenvalues put the most discriminative directions first.

25. When implementing LDA manually, what is the most likely issue if S_W is singular?

- A) Labels disappear
- B) Matrix inversion fails
- C) Data becomes nonlinear
- D) Eigenvalues become complex automatically

Answer: B) A singular S_W cannot be directly inverted; use regularization, a pseudoinverse, or dimensionality reduction.